Introduction to Computer Science and Programming

Dynamic allocation and Dynamic arrays

Lecture 16, March 30, 2026

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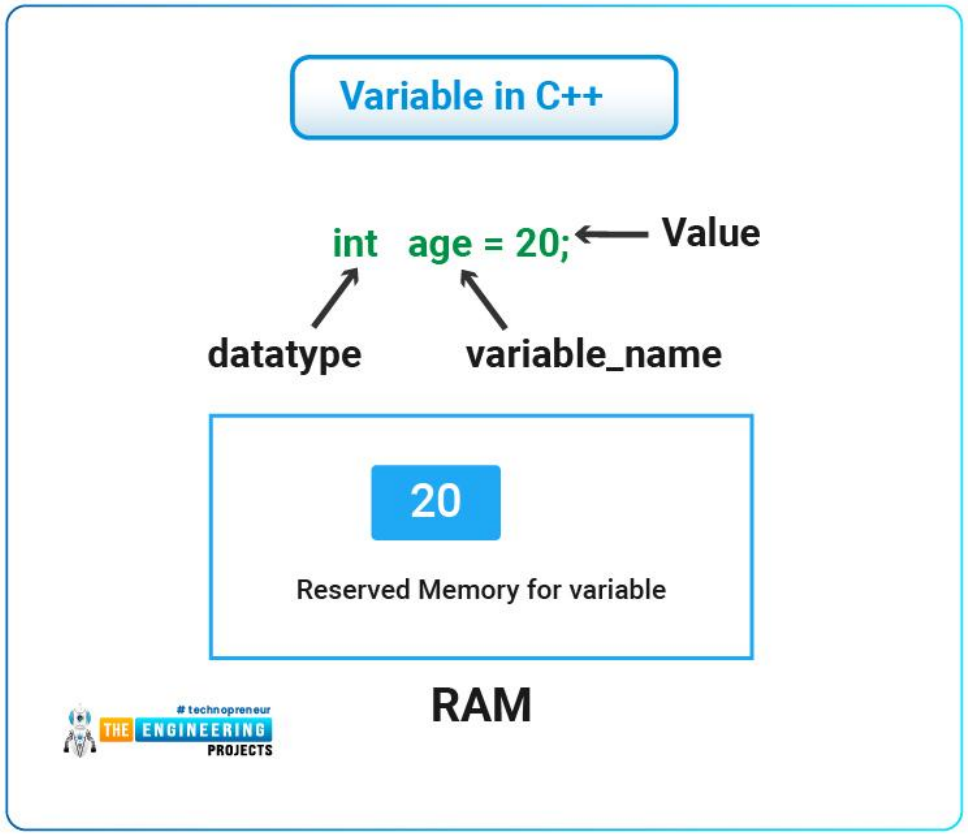
Slides based on material provided by the Polytechnic University of Turin (C++ course) and IP course (University of Genova)

Dynamic allocation

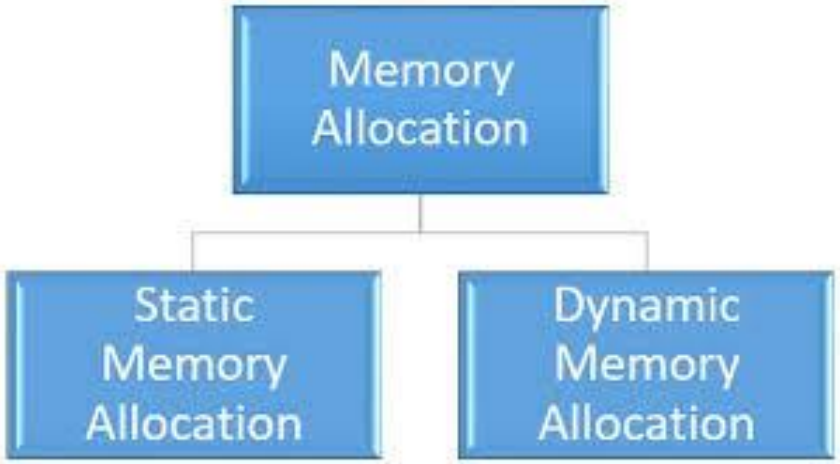
What Is Memory Allocation?

- Memory allocation is the **process of reserving memory for variables and data**
- It allows a program to store values, objects, and arrays in memory
- In C++, memory can be allocated in two main ways:
  - **Static allocation**: memory size is fixed in advance (compile time)
  - **Dynamic allocation**: memory is requested at runtime as needed
- Proper memory allocation is important for:
  - storing data efficiently
  - avoiding waste of memory
  - handling variable-size data structures



Static allocation

- **Static allocation** refers to the process of reserving memory at **compile time**, before the program starts executing
- The memory required for data structures is determined during the compilation, and **it remains fixed** for the duration of the execution
- This means that the size and number of these allocated elements **cannot change dynamically during runtime**
- In contrast, **dynamic allocation** would allow you to allocate memory during runtime



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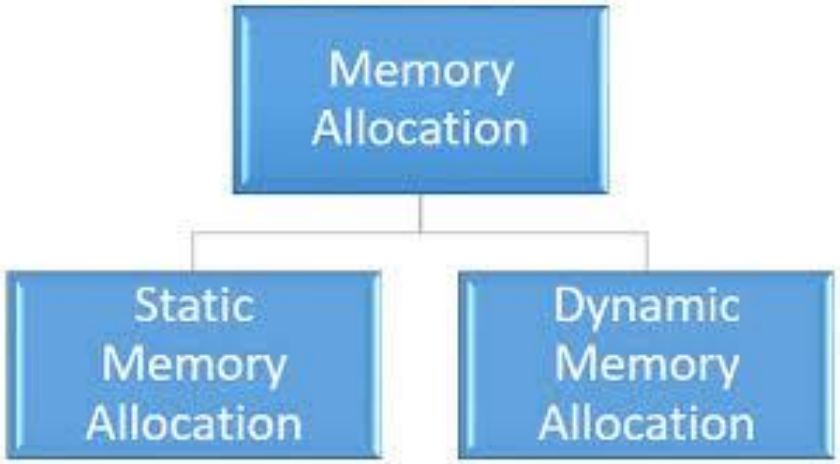

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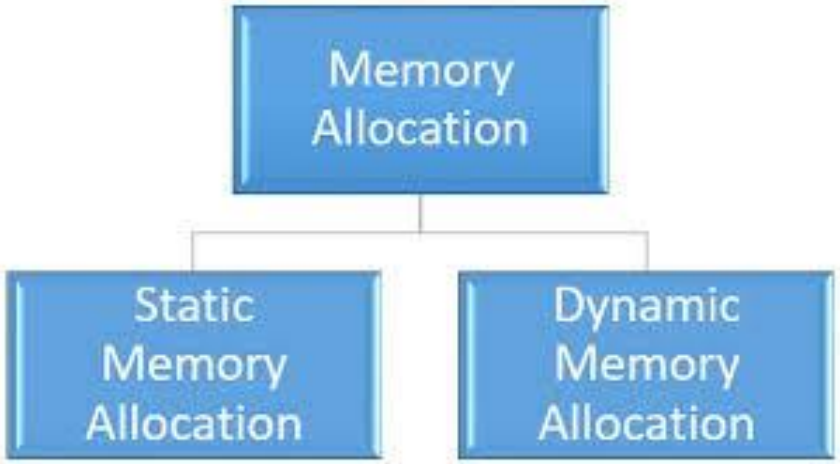

- Memory is reserved at compile time
- Memory allocated is fixed
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- Memory is reserved at run time
- Memory allocated is not fixed
- Size can change

Static allocation example

```
int main() {
    int numbers[5] = {10, 20, 30, 40, 50};  // static allocation

    for (int i = 0; i < 5; i++) {
        cout << numbers[i] << " ";
    }

    return 0;
}
```

- ‘numbers’ is an array of fixed size 5 (**static** array)
- Its size is known before execution (**compile time**)
- The size of the array can not change during execution (**fixed**)
- Memory is allocated automatically (the compiler allocates it)

Cases where static allocation is fine

- Static arrays are good when the number of elements is **fixed and known in advance**
  - **We know the size and it won't change!!!**
- Days of the week

```
string days[7] = {"Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"};
```

- Months of the year

```
string months[12] = {"Jan", "Feb", "Mar", "Apr", "May", "Jun",
                     "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};
```

- A small board with fixed size

```
char ticTacToe[3][3];
```

Problems with static allocation

- The main problem with **static allocation** is that the size is **fixed in advance**. This makes it simple, but also **inflexible**
  - With static allocation, you usually need to know the size in advance
  - If you allocate **more space** than you really need, some memory is **unused**
  - If you allocate **too little space**, you **cannot store more data** safely
  - Large statically allocated arrays inside a function may use too much stack memory
    - Overflow the stack
    - Crash!

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;

    // int arr[n];   // not standard C++
    return 0;
}
```

E.g., we need to store students in an array, but we do not know the number in advance

Motivation (static vs. dynamic)

- In many practical cases, we need to manipulate **dynamic structures**
  - e.g., when we do not know in advance the size that a given structure may reach
    - E.g., Number of students entered by the user
- The most effective choice is to allow the **program to dynamically allocate the necessary memory**

- Two options:
  - **Using the command 'new'**
    - This means the programmer **manually requests memory** at runtime
    - E.g., **Dynamic arrays**
  - **Automatically**
    - This usually means memory is managed **automatically by the language or by objects**
    - E.g., **std::vector**

Pointers and dynamic allocation

- The use of pointers so far (pointing to statically allocated variables) is not very interesting …
- However, pointers are essential for **dynamic memory management**
- Without a pointer, you would have **no way to refer to that memory after allocation**
- Pointers let the program keep track of where that memory is

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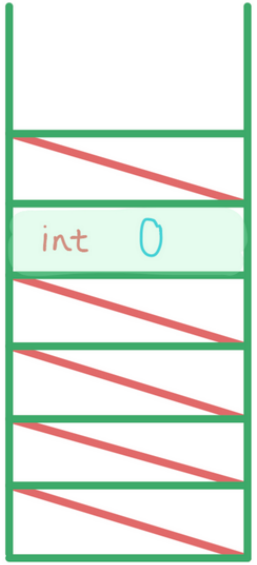

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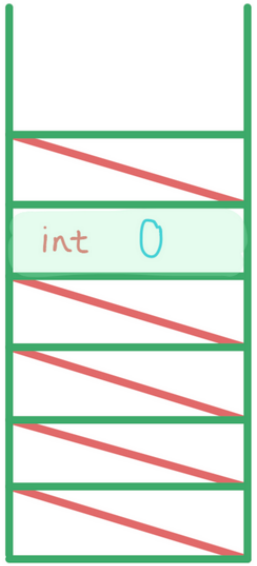

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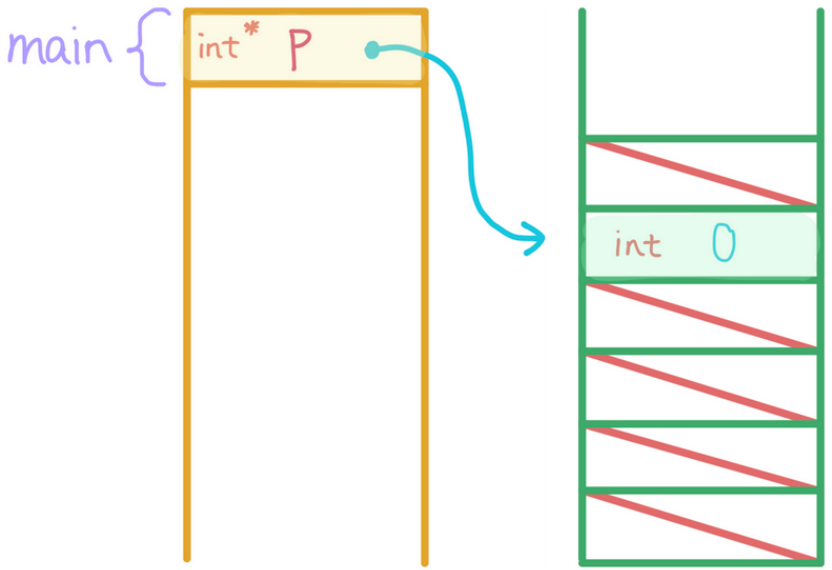
Using a pointer!!!



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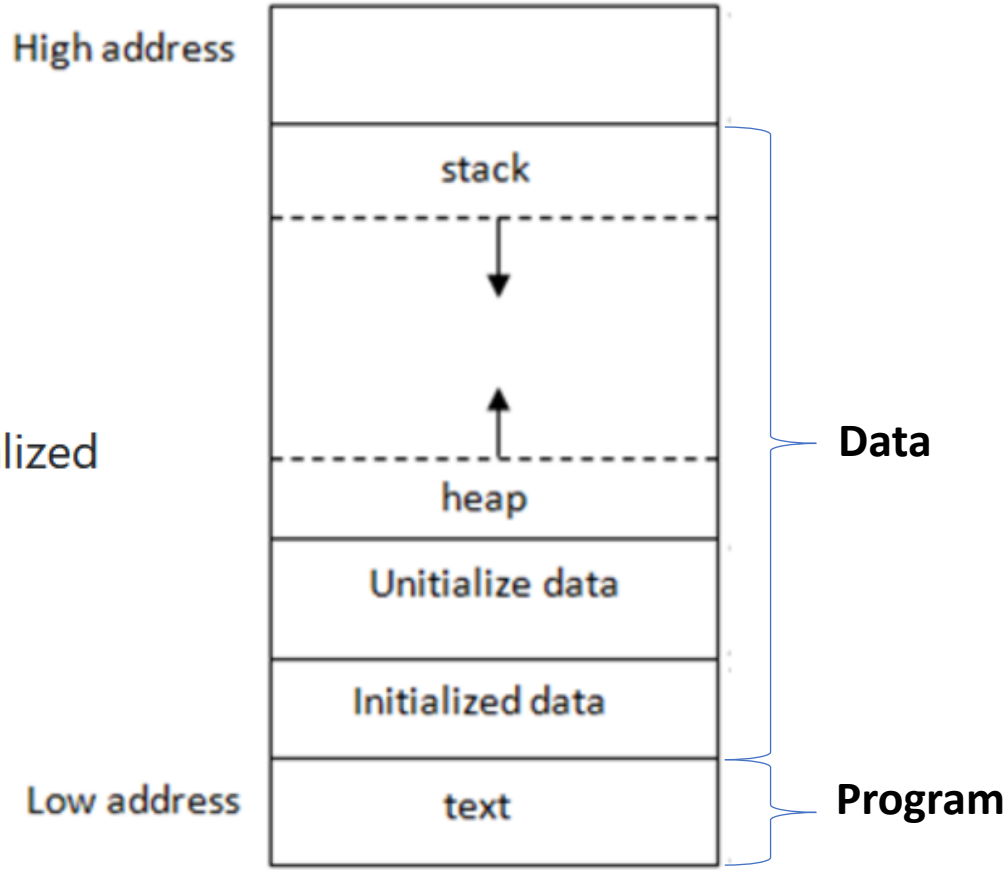
How can we refer to it?

Already introduced in Lecture 12 about Functions

Memory layout of C++ program

- When it is running, a C++ program has its own **memory layout**, separated from other programs

- `stack`: stores local variables
- `heap`: dynamic memory for programmer to allocate
- `data`: stores global variables, separated into initialized and uninitialized
- `text`: stores the code being executed



https://courses.grainger.illinois.edu/cs225/sp2023/resources/stack-heap/

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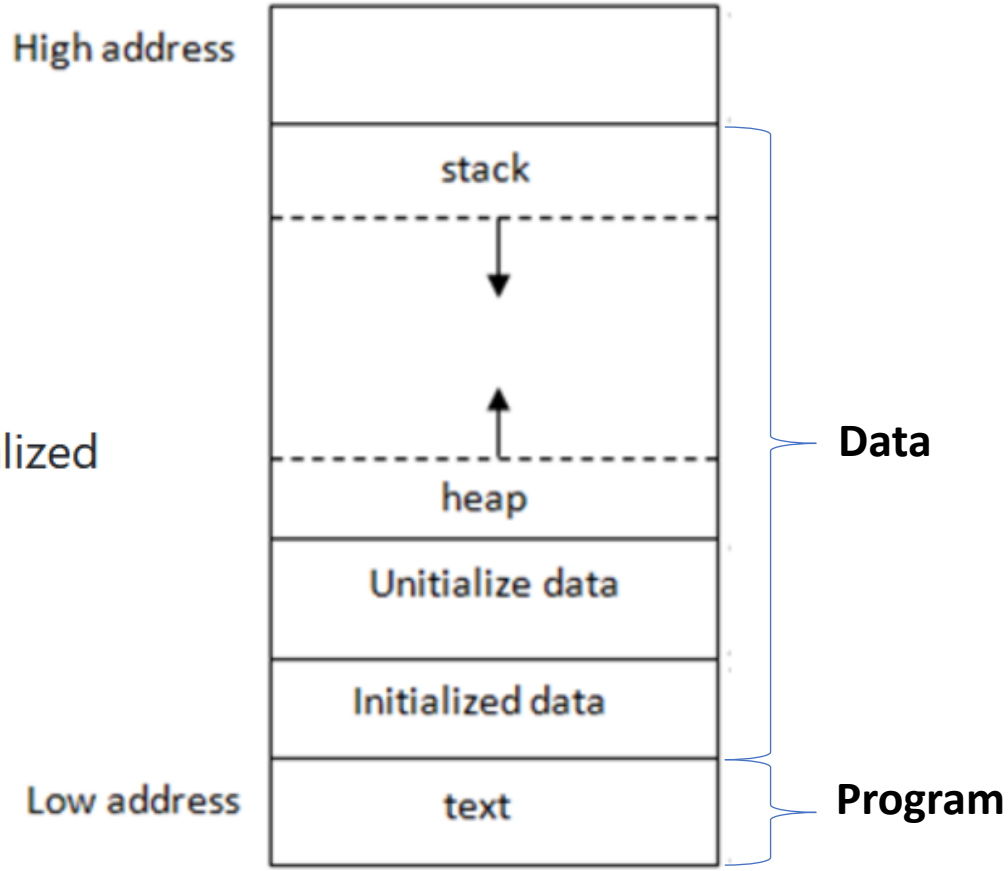

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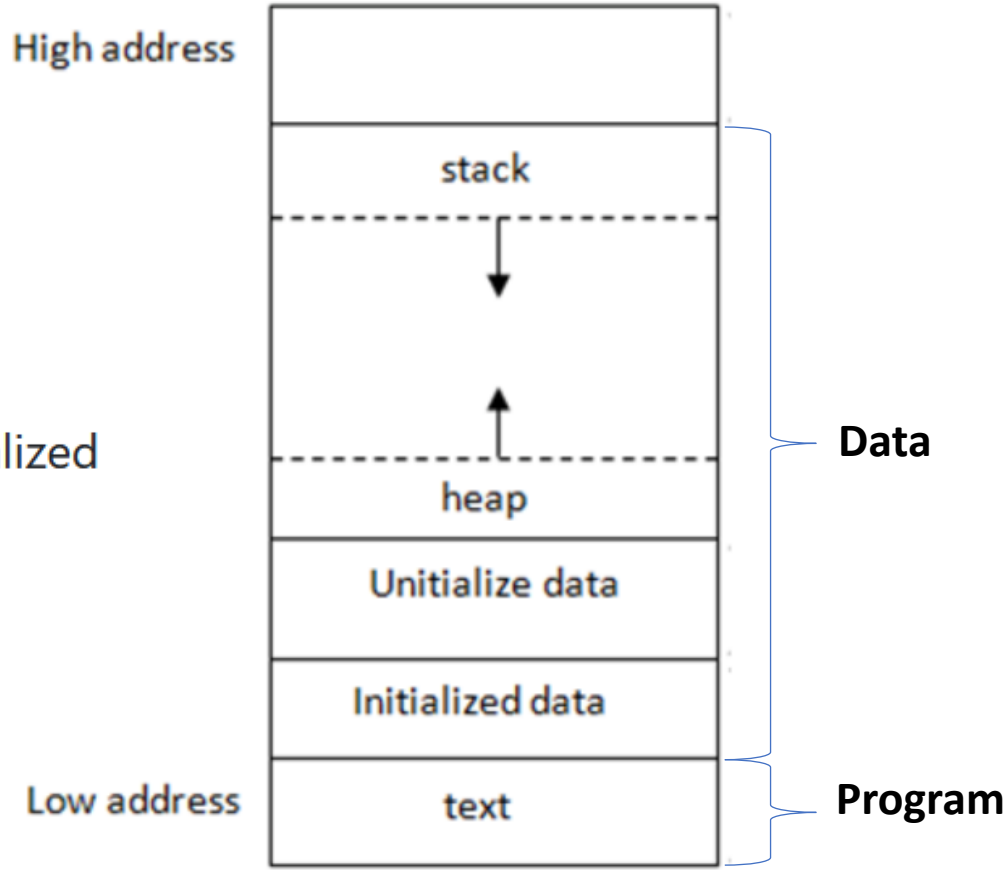

Dynamic structures are located in the heap!

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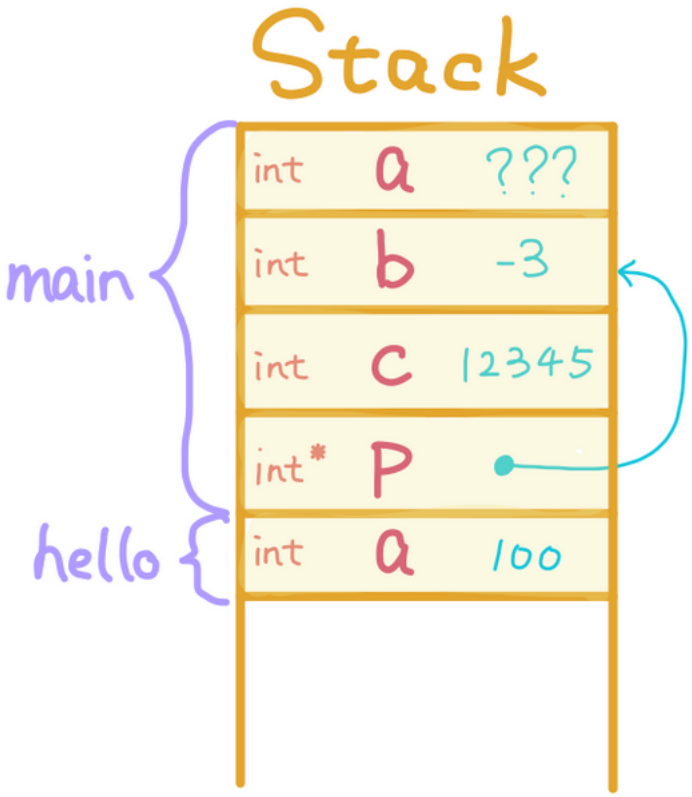
Stack

- Every **time a function is called, the machine allocates some stack memory** for it
  - Activation record
- When a **new local variables is declared, more stack memory is allocated** for that function to store the variable
- Such allocations make the stack grow downwards
- **The allocation and deallocation for stack memory is automatically done**



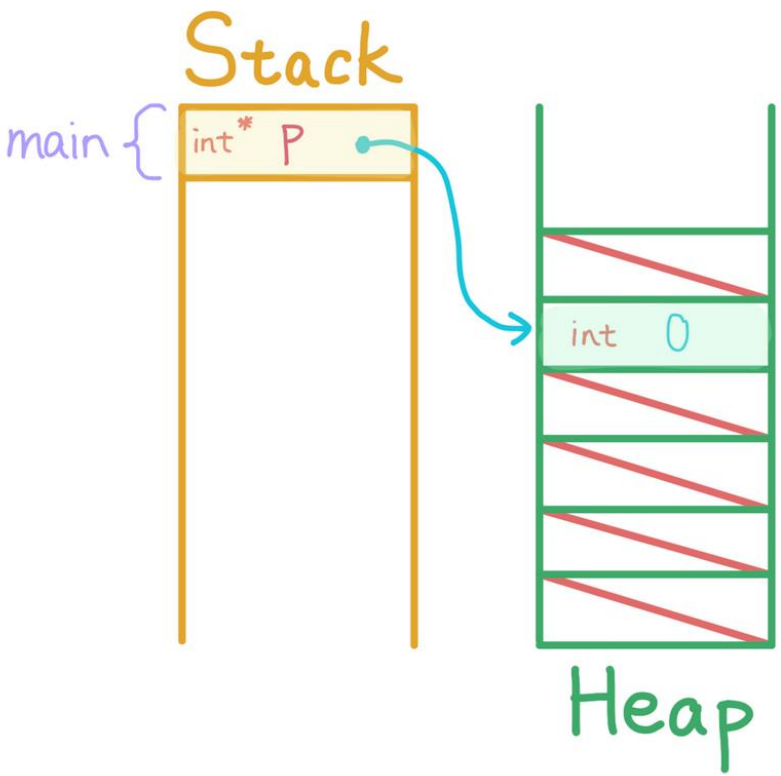

```
int hello() {
  int a = 100;
  return a;
}
int main() {
  int a;
  int b = -3;
  int c = 12345;
  int *p = &b;
  int d = hello();
  return 0;
}
```

The stack grows and shrinks automatically during the execution of the program

Heap

- Unlike stack memory, heap memory is **allocated explicitly by programmers,** and it won't be deallocated until it is explicitly freed

- To allocate heap memory in C++, use the keyword '**new**' followed of what you want to allocate
  - In our example **int**

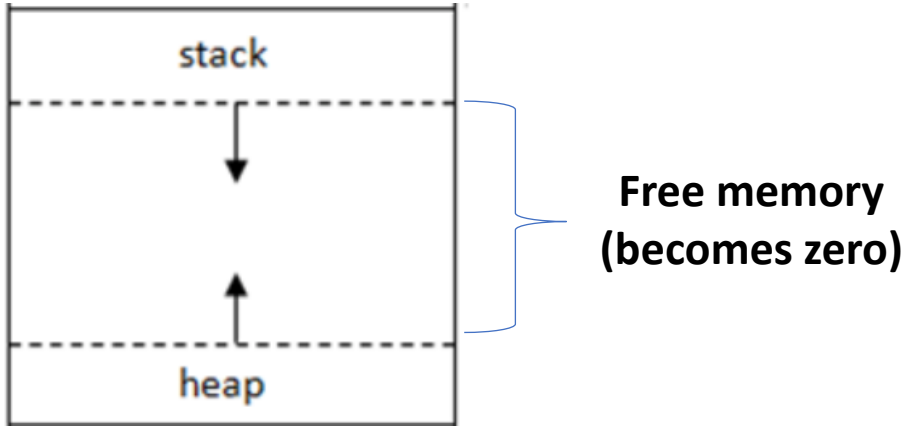



```
int main() {
  int *p = new int;
  Cube *c1 = new Cube();
  Cube *c2 = c1;
  c2->setLength( 10 );
  return 0;
}
```

P is a pointer to a memory's cell

What Happens When the Stack and Heap Collide

- **If the stack grows downward** (due to deeper function calls, large local variables, or recursion), **and the heap grows upward** (due to dynamic memory allocation), **they can eventually collide** if the program uses too much memory in either region



- This collision could cause a **segmentation fault**, a type of error in which the program **tries to access memory it isn't allowed to**, typically resulting in the **program crashing**

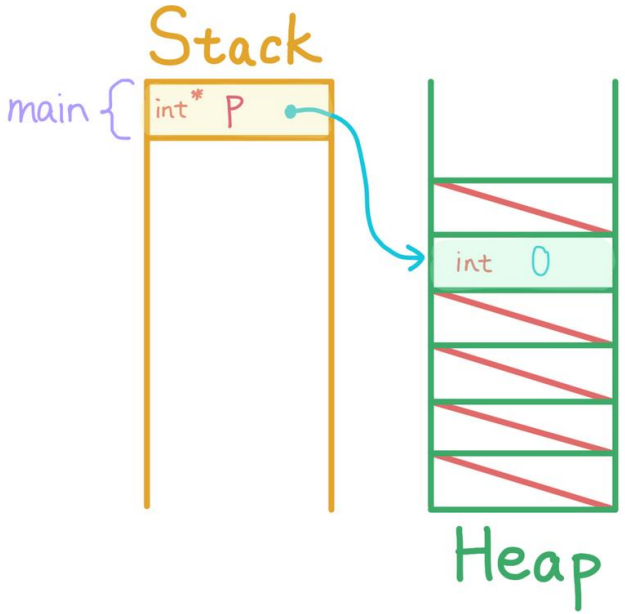
```
[01:18:40] Using gulpfile ~/Sites/sage-test/gulpfile.js
[01:18:40] Starting 'clean'...
[01:18:40] Finished 'clean' after 3.44 ms
[01:18:40] Starting 'default'...
[01:18:40] Starting 'build'...
[01:18:40] Starting 'wiredep'...
[01:18:40] Finished 'default' after 110 ms
[01:18:40] Finished 'wiredep' after 139 ms
[01:18:40] Starting 'styles'...
Segmentation fault: 11
Daenerys:sage-test Michael$
```

Heap manager

- I request the **heap manager** to allocate memory necessary to store a variable of a certain type
- The manager allocates the required memory on the heap and **returns the address**
- In C++, this operation is performed using the **new command**:

```
T *p;
p = new T;
```

- This instruction **allocates a variable of type T** on the heap and stores its address in p

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Dynamic allocation (visual representation)

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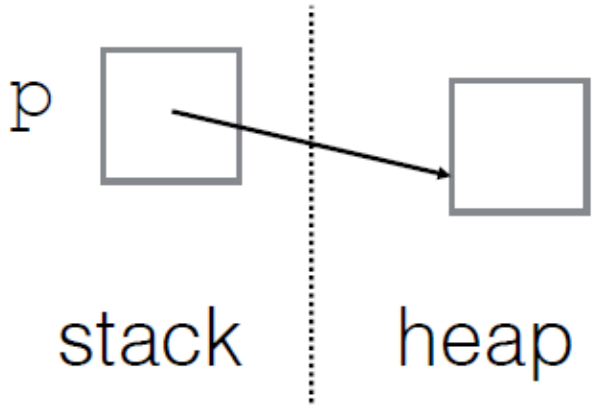

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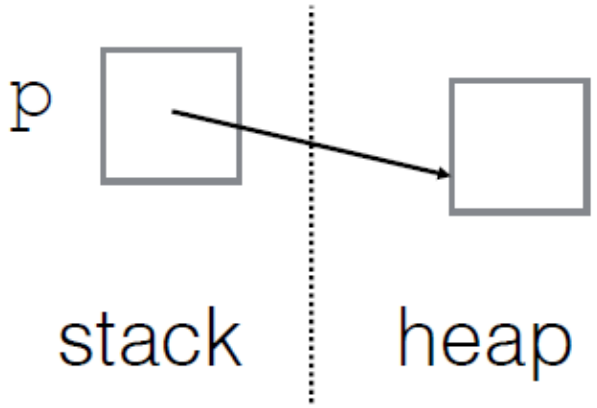

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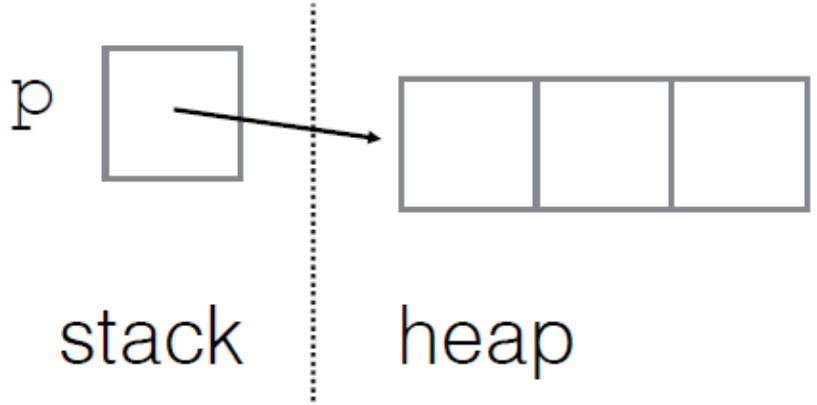
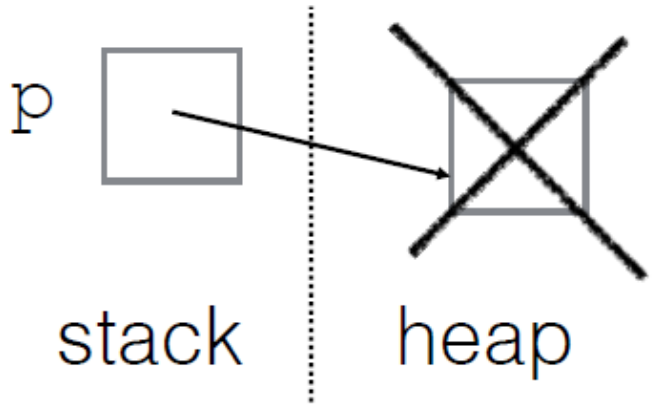
- `int size = 3;`
  `p = new T[size];`



Dynamic de-allocation

Returns a portion of memory to the heap that is no longer needed

- This operation invalidates access to that memory portion and makes it available to the heap manager for future use



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- If we have a structure with multiple elements:

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delete [] p;
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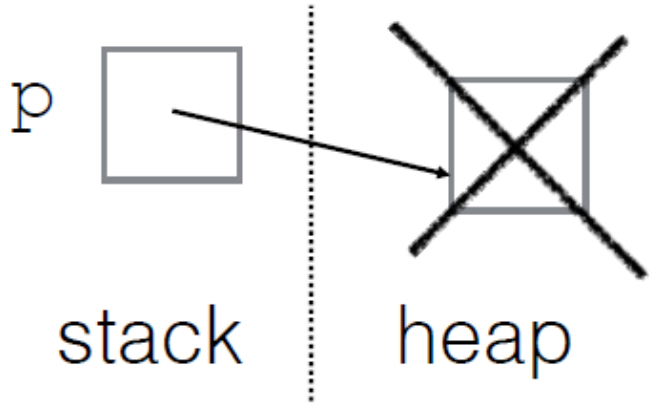

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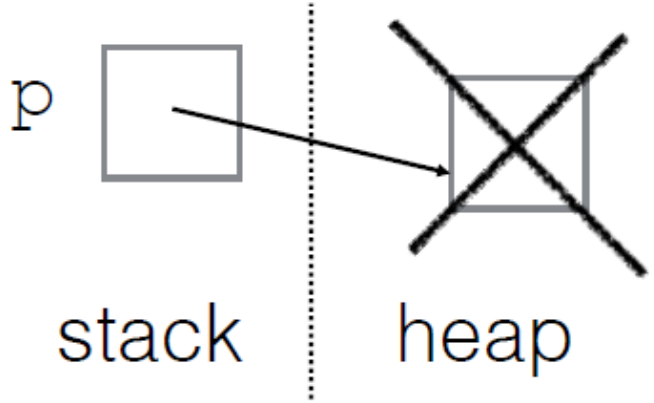


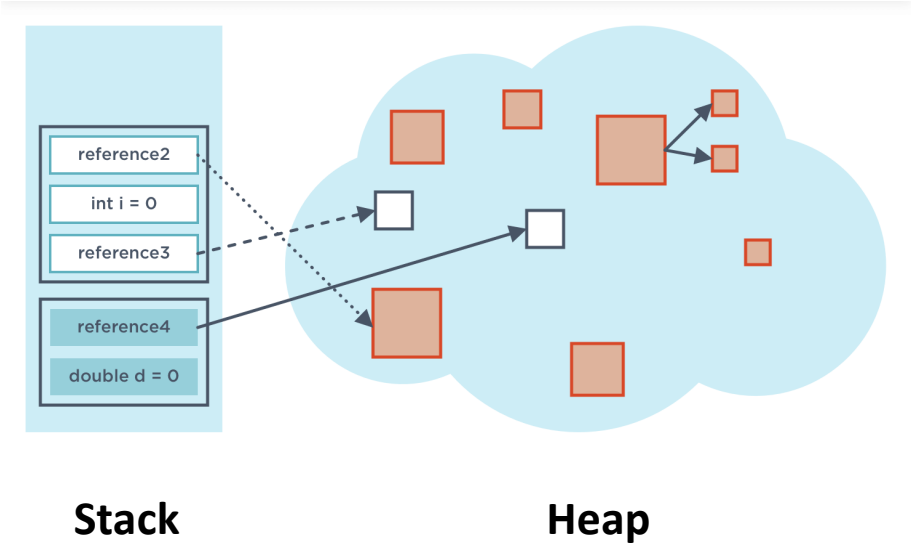
After delete, the pointer still contains the old address, but that address is no longer safe to use

Problems with Dynamic allocation

- Pay attention to:
  - **Dangling pointers**
    - A **dangling pointer** happens when a pointer still points to memory that has already been deallocated

Problems with Dynamic allocation

- Pay attention to:
  - **Dangling pointers**
    - A **dangling pointer** happens when a pointer still points to memory that has already been deallocated
  - **Memory leak**
    - A **memory leak** happens when a program allocates memory dynamically, but **never releases it**



Dangling Pointers problems

- Dangling pointers cause **undefined behavior** because they point to memory that is **no longer valid**

- Main problems:
  - **crashes**
    the **program may crash** when trying to access freed memory

  - **wrong values**
    the pointer may **read garbage** or unpredictable data

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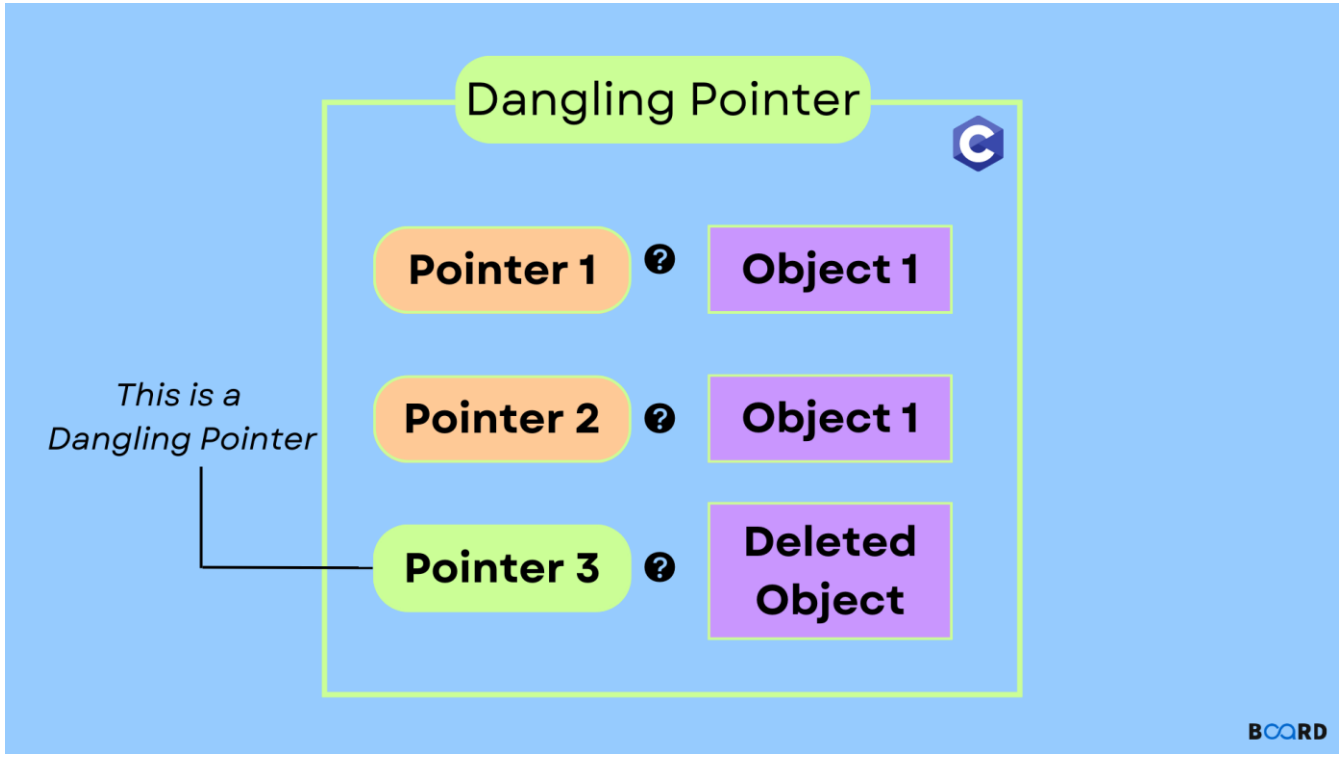
```
#include <iostream>
using namespace std;

int main() {
    int* ptr = new int;
    *ptr = 25;

    delete ptr;   // memory is released

    cout << *ptr << endl;   // dangling pointer: undefined behavior

    return 0;
}
```

Dangling pointers

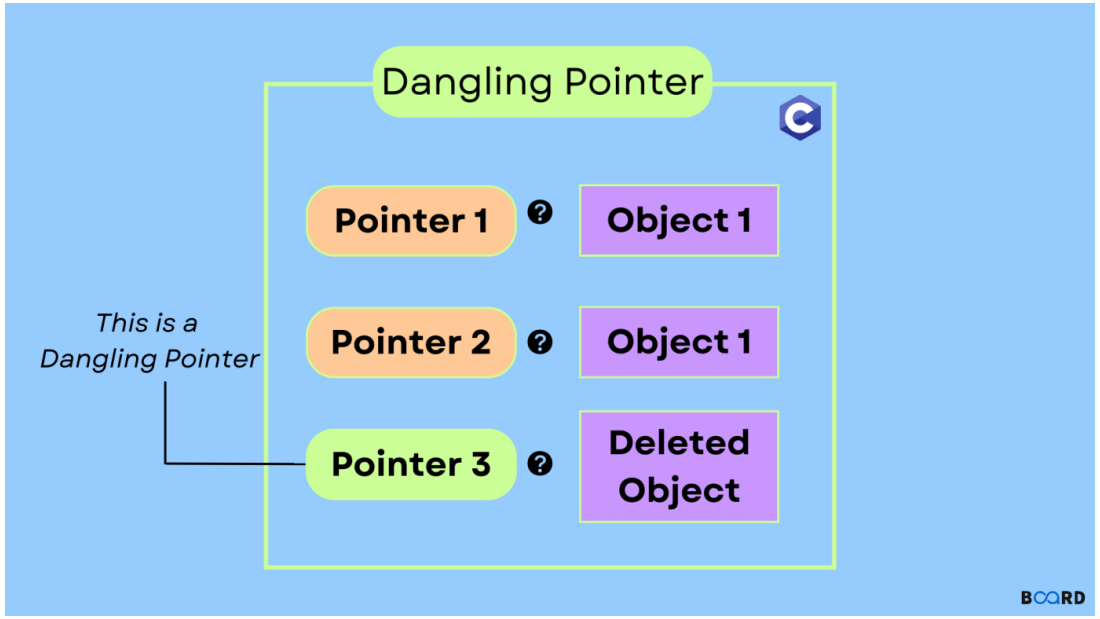


- To prevent a pointer from pointing to nothing (anymore), after a delete, it is advisable to:

```
delete [] p1;
p1=nullptr;
```

Exercise 1: Dangling pointers

- Write a simple C++ program that contains a dangling pointer
- Try to print the value associated with a dangling pointer



Memory Leak

- The **new** operator dynamically allocates memory
- When the memory is no longer needed, it should be released using **delete**
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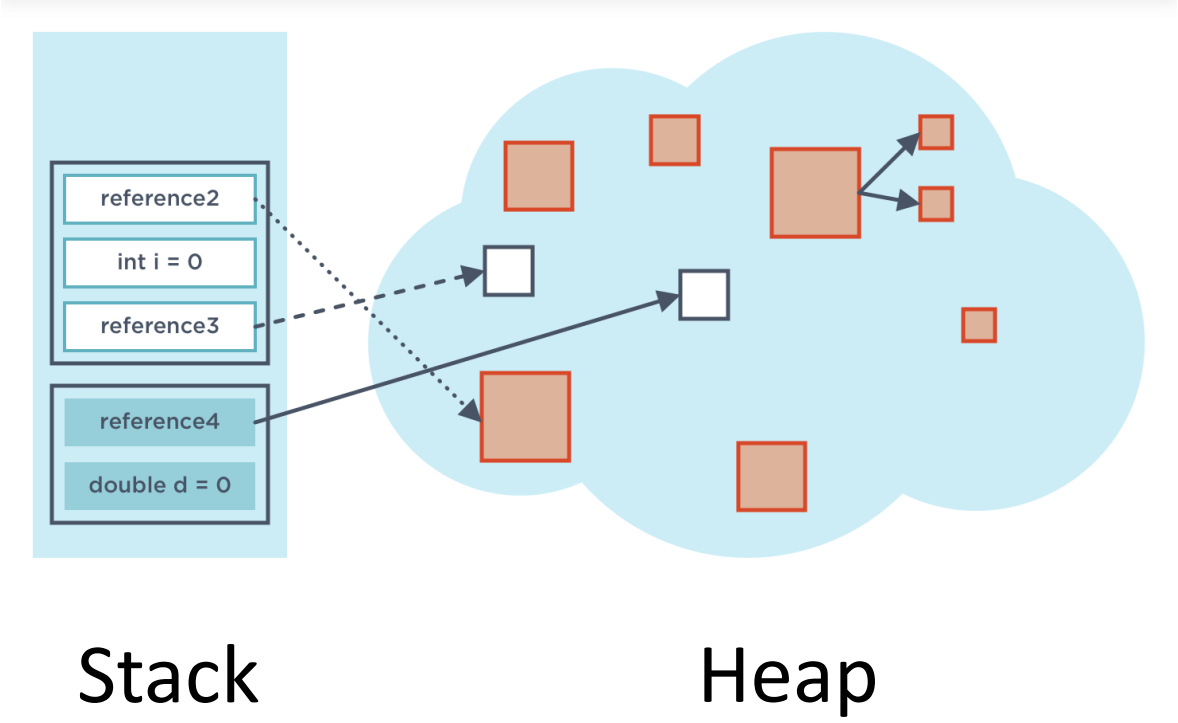

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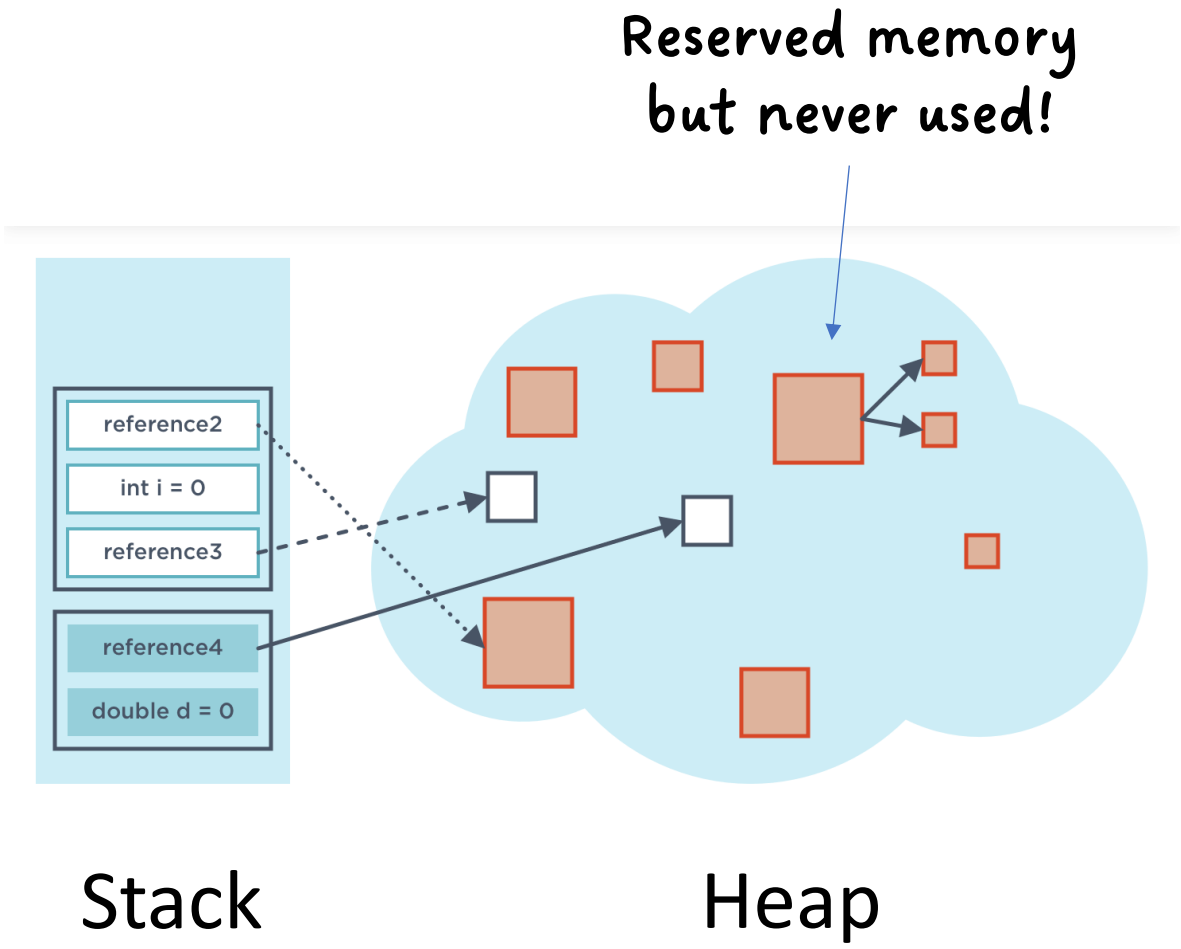


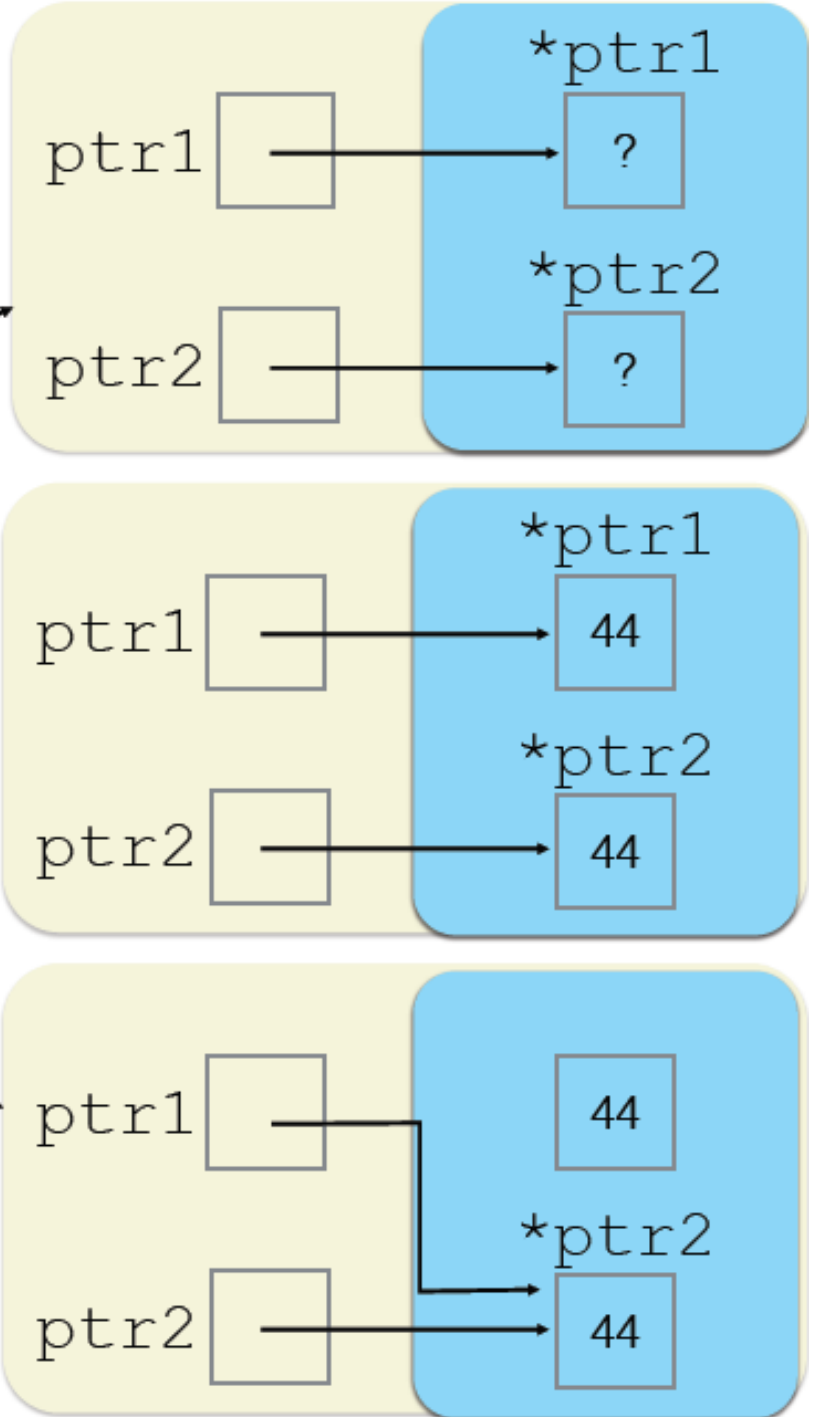
Memory Leak example

```
int *ptr1 = new int;
int *ptr2 = new int;

*ptr2 = 44;
*ptr1 = *ptr2;

ptr1 = ptr2;

delete ptr2;
ptr1=nullptr;
ptr2=nullptr;
```



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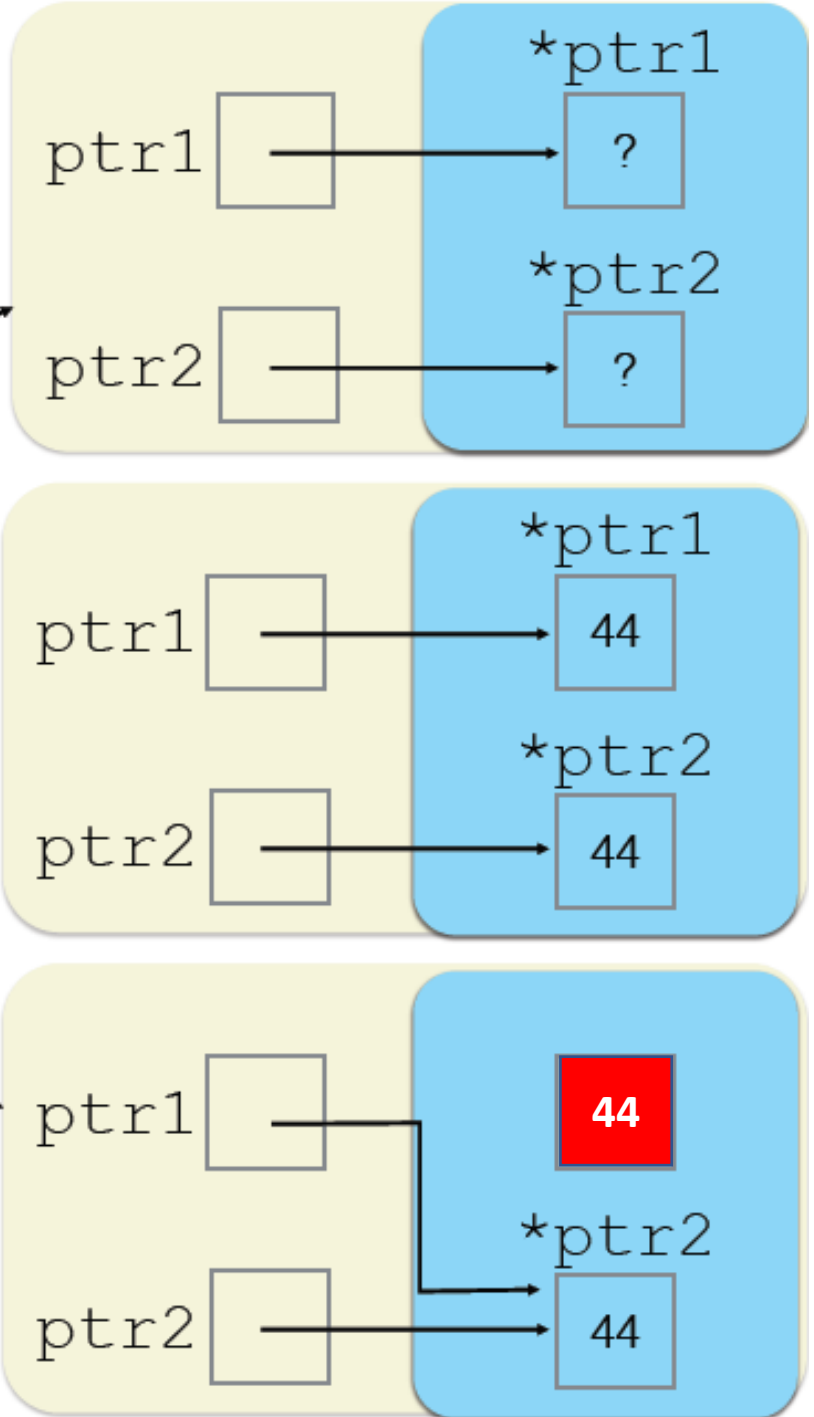

We forgot to delete ptr1, so that memory cell remains allocated

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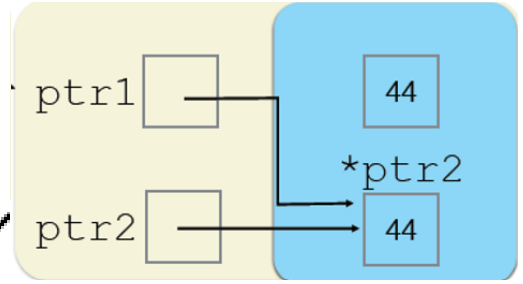

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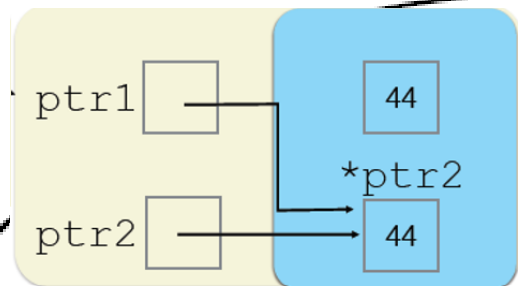
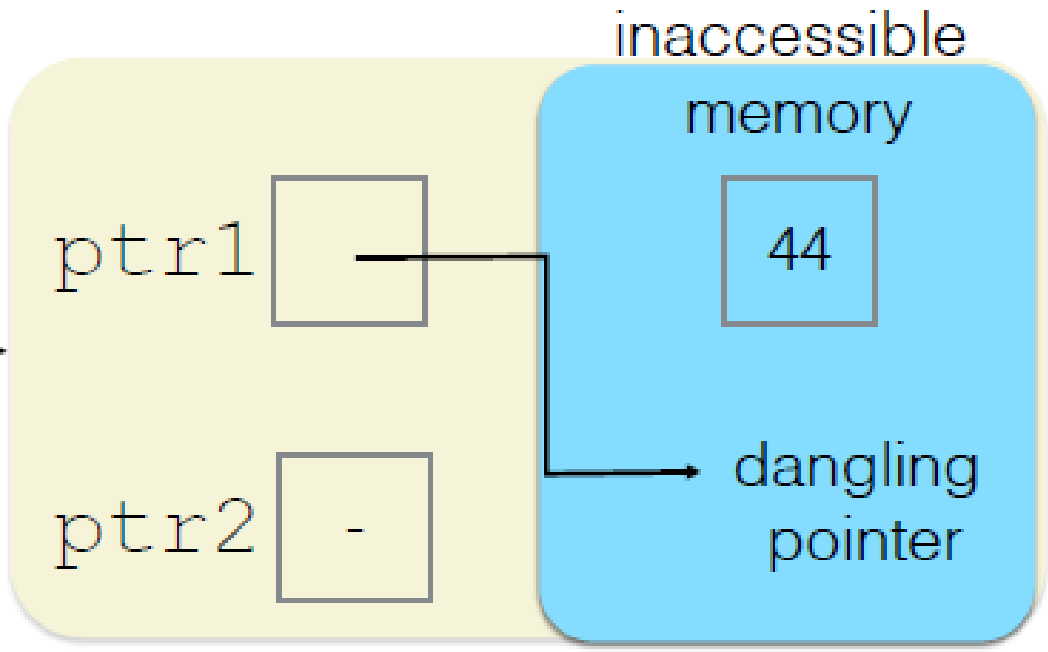
```
int *ptr1 = new int;
int *ptr2 = new int;


*ptr2 = 44;
*ptr1 = *ptr2;


ptr1 = ptr2;


delete ptr2;
ptr2 = nullptr;
```

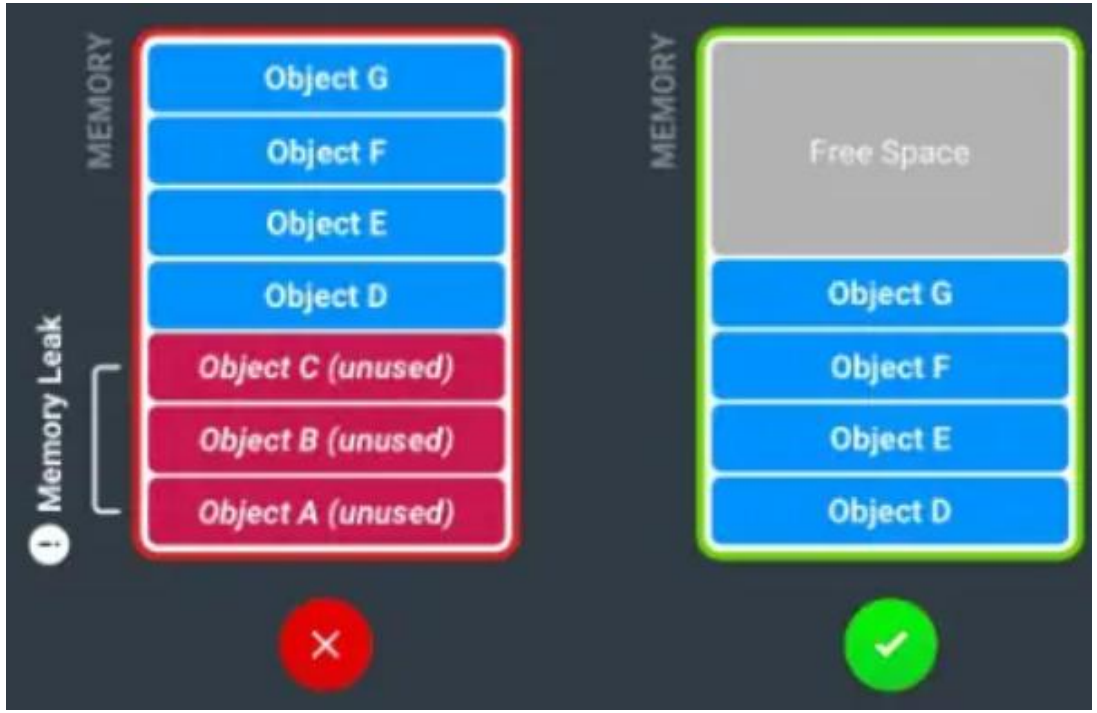




Memory Leak problems

- In large programs, memory leaks can build up over time, use more and more memory, **slow the program** down, and may eventually cause it to **crash**

- Tools like **Valgrind** or **Visual Studio's built-in memory profiling tools** can help detect and debug memory leaks
  - Advanced tools!
  - Not part of this course



All available memory is consumed!

Simple Strategy to Detect a Memory Leak

- Use a **counter** to track dynamic memory operations
- Every time new is called, the counter **increases** (**+1**)
- Every time delete is called, the counter **decreases** (**-1**)
- At the end of the program:
  - if the counter is 0, all allocated memory was released
  - if the counter is not 0, some memory was not released

- **Takeaway:**
  If the final counter value is not zero, the program may contain a **memory leak**

```cpp
#include <iostream>
using namespace std;

int counter = 0;      // Global counter to track active memory allocations

void* operator new(size_t size) {   // Redefines the new operator
    counter++;                      // Increases the counter when memory is allocated
    return malloc(size);            // Allocates raw memory and returns its address
}

void operator delete(void* ptr) noexcept {   // Redefines the delete operator
    counter--;                               // Decreases the counter when memory is released
    free(ptr);                               // Frees the allocated memory
}

int main() {
    int* p1 = new int;        // Allocates memory for one integer
    int* p2 = new int;        // Allocates memory for another integer

    delete p1;                // Frees the memory pointed to by p1
    // delete p2;            // This line is commented out, so p2 is not freed

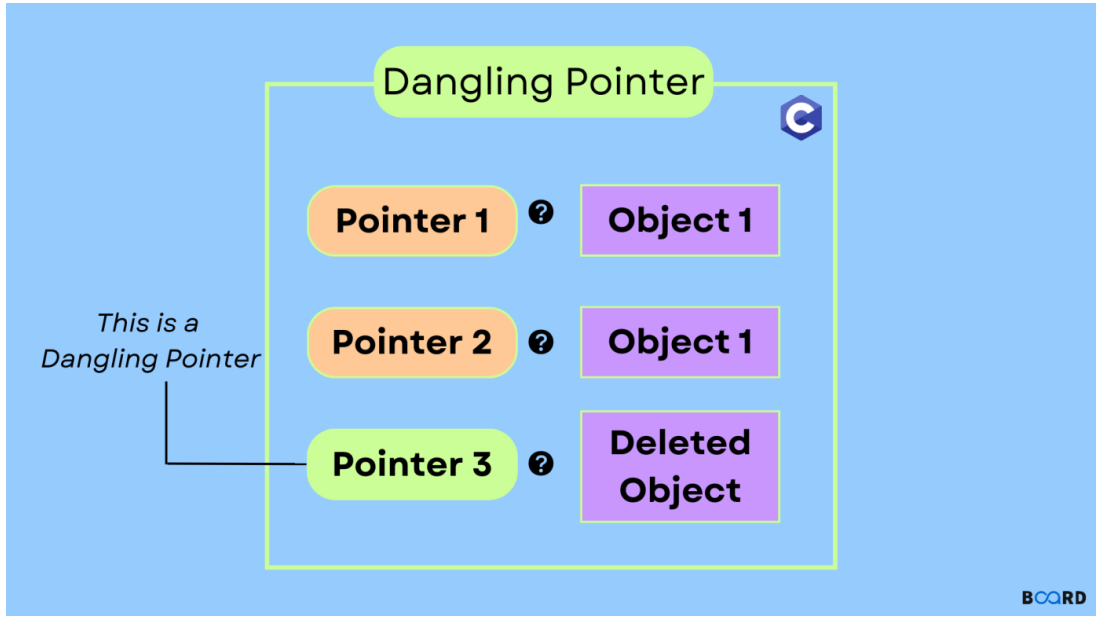
    cout << "Allocations not freed: " << counter << endl;   // Prints how many allocations were not released

    return 0;
}
```

This code uses advanced C++ concepts, such as operator overloading, which we will cover in the final lectures

Exercise 2: Memory leak

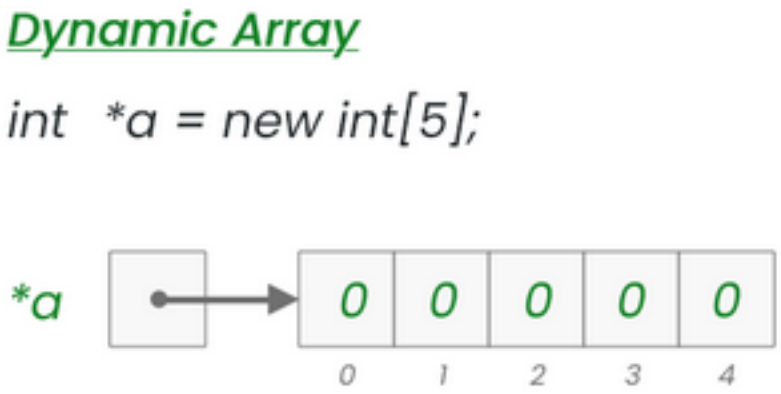
- Write a simple C++ program that contains Memory leaks
- Try to execute the program given in the previous slide



Dynamic arrays

Dynamic array

- A **dynamic array** in C++ is an array whose **size can be determined at runtime**, as opposed to a static array, where the size is fixed at compile-time
- Dynamic arrays are **useful when the size of the array is not known in advance** or may change during the execution of the program
- In C++, dynamic arrays are created **using pointers** and the new keyword for memory allocation and delete or delete[] for deallocation



Dynamic arrays vs static arrays

```
int a[10];
int *p;
p = new int[10];
```

Static array

Dinamic array

a	p
resides in the stack	resides in the heap
fixed size **Compile time**	variable size **Run time**
static binding between name and array (*a* is a **constant pointer**)	dynamic binding between name and array

Dynamic arrays vs static arrays

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int a[10];
```

Static array

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int *p;
p = new int[10];
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Dynamic arrays vs static arrays

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Static array

```
int *p;
p = new int[10];
```

Dinamic array

a	p
resides in the stack	resides in the heap
fixed size **Compile time**	variable size **Run time**
static binding between name and array (*a* is a **constant pointer**)	dynamic binding between name and array

p is a real pointer variable, so it can later point somewhere else

Static Arrays vs Variable-Size Dynamic Arrays

- Static arrays require a **size known before execution**, while dynamic arrays can be created with a **size determined at runtime**

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;

    int arr[n];   // compile-time error in standard C++
    return 0;
}
```

Static array

Static Arrays vs Variable-Size Dynamic Arrays

- Static arrays require a **size known before execution**, while dynamic arrays can be created with a **size determined at runtime**

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;

    int arr[n];   // compile-time error in standard C++
    return 0;
}
```

Static array

Dynamic array

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;

    int* arr = new int[n];   // size decided at runtime

    delete[] arr;
    return 0;
}
```

Dynamic arrays and access to elements

- **Access to elements can be performed as with static arrays** (including pointer arithmetic)

```
int *p;
p = new int[5];
for (int i=0;i<5;++i)
        p[i]=i*i;
```

```
int *p;
p = new int[5];
for (int i=0;i<5;++i)
        *(p+i)=i*i;
```

Dynamic arrays and access to elements

- **Access to elements can be performed as with static arrays** (including pointer arithmetic)

```
int *p;
p = new int[5];
for (int i=0;i<5;++i)
        p[i]=i*i;
```

```
int *p;
p = new int[5];
for (int i=0;i<5;++i)
        *(p+i)=i*i;
```

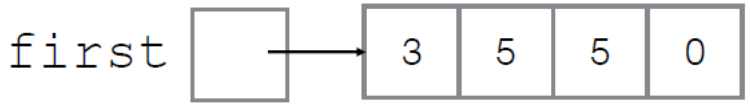
- Just like in the static case, **accessing memory outside the range** (out of range) can result in one of two scenarios:
  - If the cell is outside the memory allocated by the heap, the **program aborts**
  - If the cell is within the allocated area, the program continues but produces **unpredictable results**

Shallow copy and deep copy

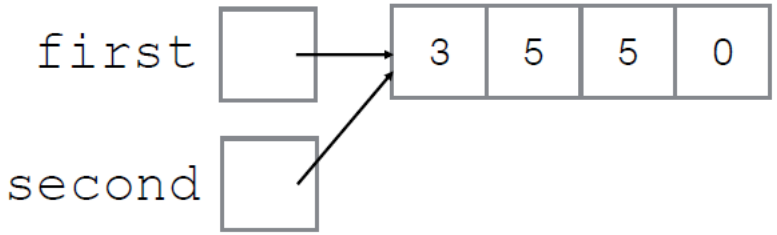
- Consider the following statements

```
int *first;
int *second;
first =  new int[10];
```

- Insert some elements in the ‘first’ array



- **Shallow copy**:
  - second = first



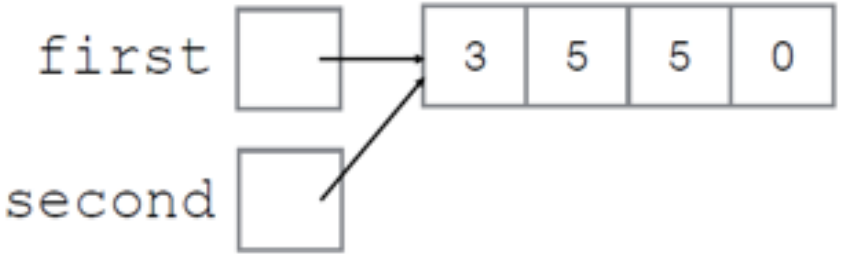
Shallow copy and deep copy

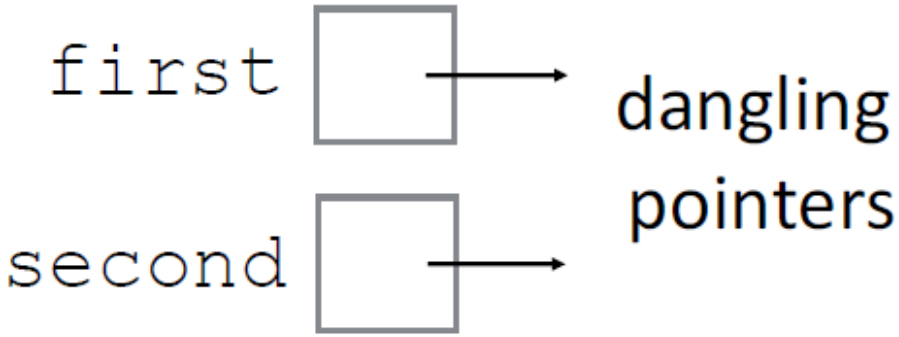
- what happens if:

```
delete [] second;
?
```

Shallow copy and deep copy

- what happens if: `delete [] second;`
  `?`

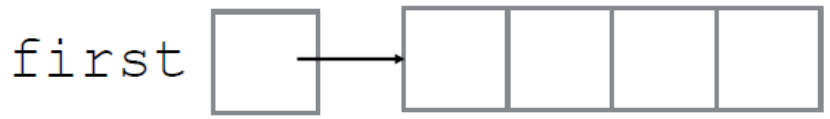




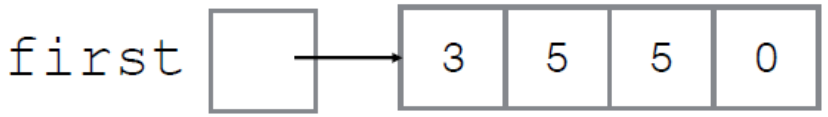
Shallow copy and deep copy

- Consider the following statements

```
int *first;
int *second;
first =  new int[10];
```

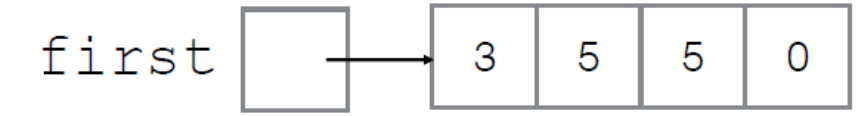


- Insert some elements in the ‘first’ array



- **Deep copy:**

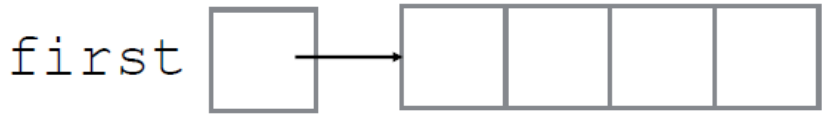
```
second = new int[10];
for (int i=0;i<10;i++)
 second[i]=first[i];
```



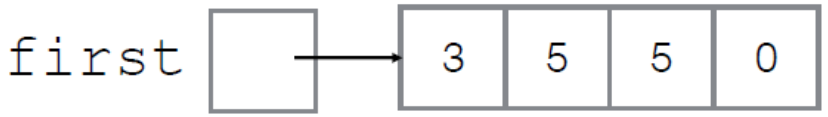
Shallow copy and deep copy

- Consider the following statements

```
int *first;
int *second;
first =  new int[10];
```



- Insert some elements in the 'first' array



- **Deep copy:**

```
second = new int[10];
for (int i=0;i<10;i++)
 second[i]=first[i];
```

Now `delete [] second;` only deletes the data associated with second

Resizing a Dynamic Array

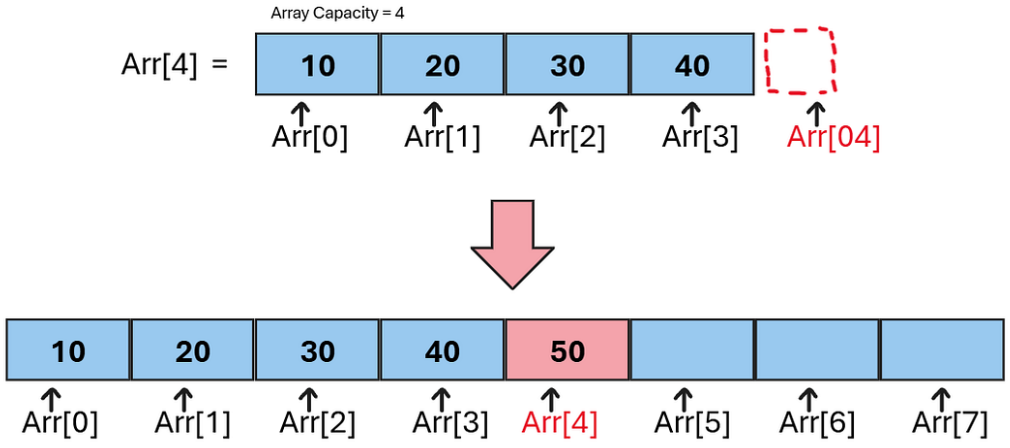
- A dynamic array **cannot grow by itself**
- For example:
  - int* p = new int[4];
- Here, the array has space for exactly **4 elements**. If you later need to store a 5th element, the existing array cannot simply expand in place

Solution:

Resize: if the current dynamic array no longer has enough space, we must create another **bigger array** and move the data into it

Basic idea:

1. Allocate a **new array** with a larger size
2. Copy the elements from the old array into the new one (**deep copy**)
3. Delete the old array
4. Make the pointer refer to the new array



We create a bigger array

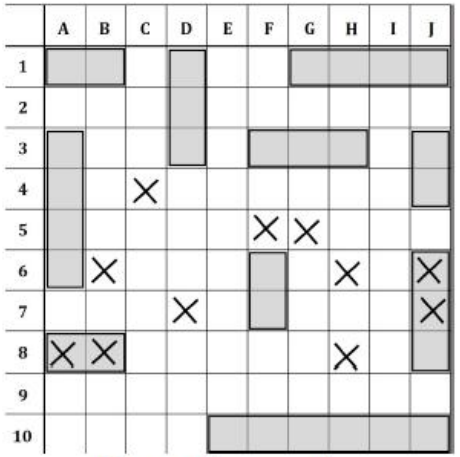
```cpp
#include <iostream>
using namespace std;

int main() {
    int size = 5;                   // it is no longer a const
    int* a = new int[size];         // allocation on the heap
    int n = 0;                      // number of inserted elements
    int value;                      // temporary variable for input

    // --- Read values into the array until 0 is entered
    while (cin >> value && value != 0) {
        if (n >= size) {                            // when the array is full
            size = size * 2;                        // double the value of size
            int* temp = new int[size];              // create a temporary array twice as large
            for (int i = 0; i < n; ++i) {
                temp[i] = a[i];                     // deep copy
            }
            delete[] a;                             // free the old (smaller) array
            a = temp;                               // update the pointer
        }

        a[n] = value;                               // store the new value in the array
        n++;                                        // increase the number of inserted elements
    }

    // --- Print what was inserted (it may be less than size!)
    for (int i = 0; i < n; ++i)
        cout << a[i] << endl;
    delete[] a;                                     // free the dynamic array before the program ends
    return 0;
}
```

Dynamic ad static arrays

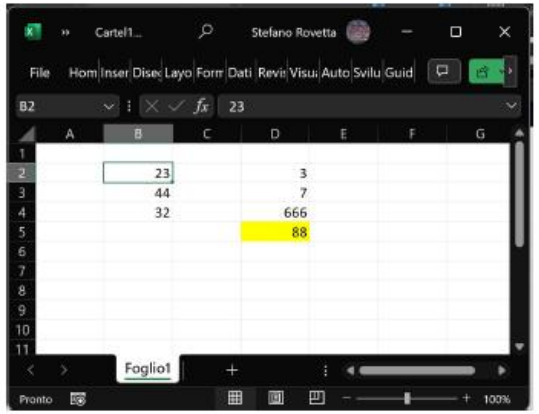
```
#include <iostream>
using namespace std;

const int SIZE = 10;   // fixed size used for both arrays

void print_array(int *a, int size) {   // prints all elements of the array
    for (int i = 0; i < size; ++i)
        std::cout << *(a + i) << " ";  // prints the value at position i
    std::cout << std::endl;            // moves to the next line
}

void init_array(int *a, int size, int K) {   // initializes all elements to K
    for (int i = 0; i < size; ++i)
        *(a + i) = K;                      // stores K in each array position
}

int main() {
    int *p1;                        // pointer to a dynamic array
    p1 = new int[SIZE];             // dynamic array allocated on the heap
    int v[SIZE];                    // static array allocated on the stack

    init_array(v, SIZE, 1);         // fills the static array with 1
    print_array(v, SIZE);           // prints the static array

    init_array(p1, SIZE, 3);        // fills the dynamic array with 3
    print_array(p1, SIZE);          // prints the dynamic array

    delete[] p1;                    // releases the dynamic array
    return 0;                       // ends the program
}
```

They are equivalent when used as parameters in a Function

Bi-dimensional arrays



Battaglia navale



Foglio di calcolo

Pixel (immagine)

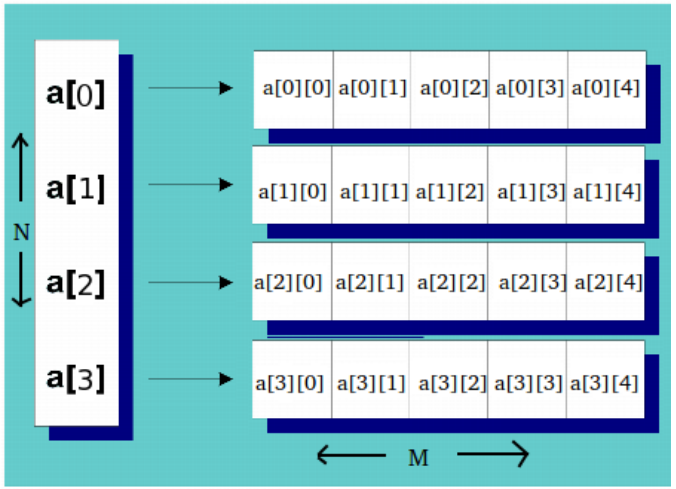
- Characterized by two indices: a 'number of rows' and a 'number of columns

Already introduced in Lecture 14

(Static) Bi-dimensional arrays

- They are arrays of arrays – Arrays whose elements are themselves arrays

```
A[i][j] = element in row i, column j
A[i]    = the entire row i (an array)
```



- **Always allocated on the stack!!!**

(Dynamic) Bi-dimensional arrays 1

- **Always allocated on the heap!**

```
int** m = new int*[rows];
```

First step in creating a **dynamic 2D array**

(Dynamic) Bi-dimensional arrays 1

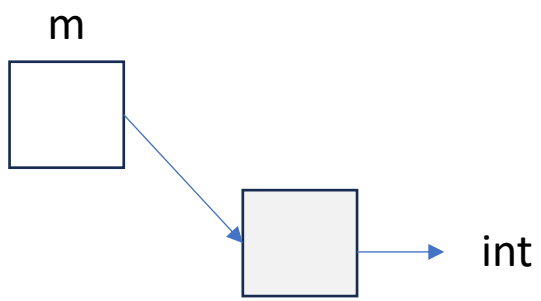
- **Always allocated on the heap!**

```
int** m = new int*[rows];
```

First step in creating a **dynamic 2D array**

- **1. int** m**

  This means:
  - m is a **pointer to a pointer to int**
  - in practice, m will store the address of an array of pointers
  - each of those pointers will later point to one row



(Dynamic) Bi-dimensional arrays 1

- **Always allocated on the heap!**

```
int** m = new int*[rows];
```

First step in creating a **dynamic 2D array**

- **1. int** m**

  This means:
  - m is a **pointer to a pointer to int**
  - in practice, m will store the address of an array of pointers
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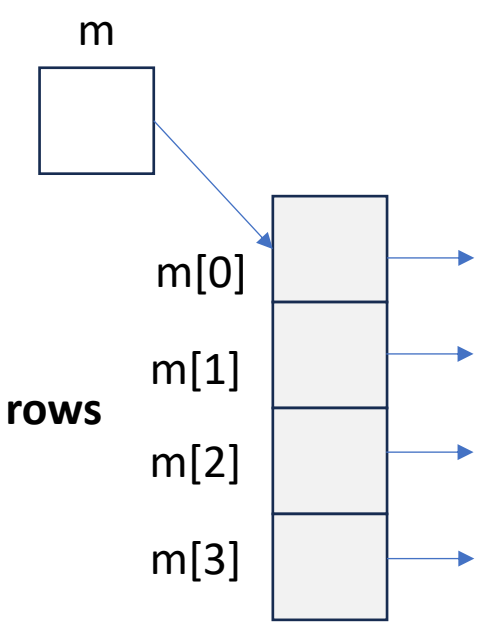


- **2. new int*[rows]**
  - This allocates memory for:
    - an array with rows elements
    - each element is of type int*
  - So this line creates: an array of **row pointers**

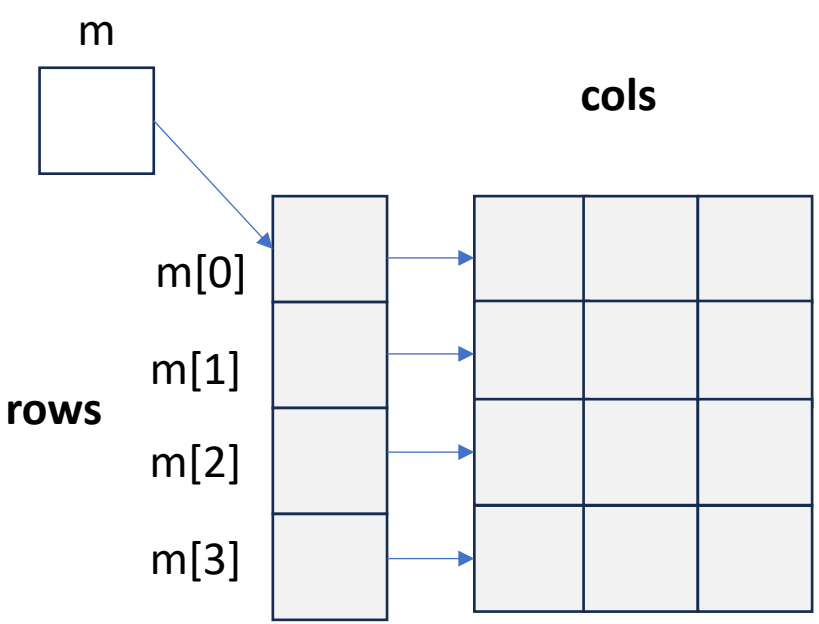
(Dynamic) Bi-dimensional arrays 2

- **3. What happens next**

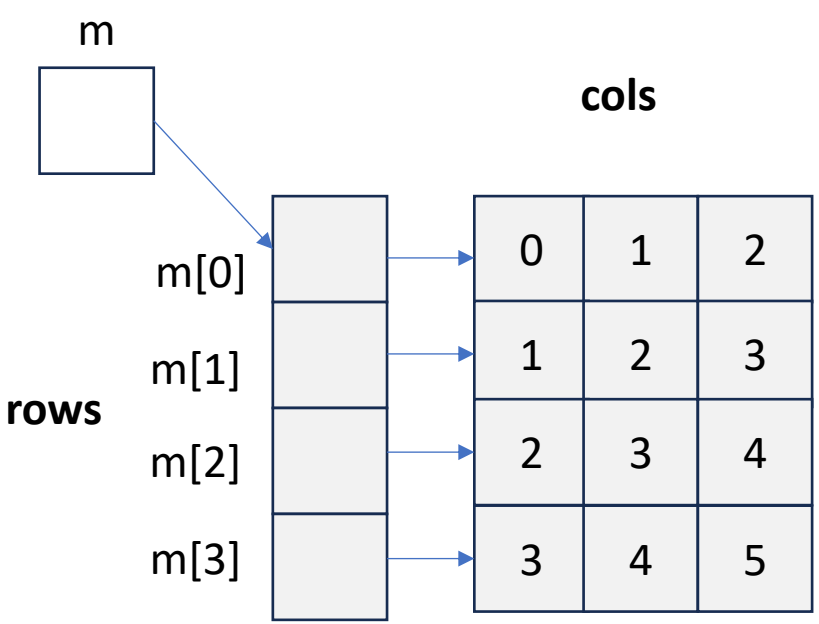
Usually after that, you allocate each row:

```
for (int i = 0; i < rows; i++) {
    m[i] = new int[cols];
}
```

Now each m[i] points to an array of cols integers



```cpp
#include <iostream>
using namespace std;

int main() {
    int rows = 4;
    int cols = 3;

    int** m = new int*[rows];          // allocate rows
    for (int i = 0; i < rows; i++) {
        m[i] = new int[cols];          // allocate columns for each row
    }

    // example initialization
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            m[i][j] = i + j;
        }
    }

   // print the matrix
   for (int i = 0; i < rows; i++) {
       for (int j = 0; j < cols; j++) {
           cout << m[i][j] << " ";
       }
       cout << endl;
   }

   // deallocation
   for (int i = 0; i < rows; i++) {
       delete[] m[i];
   }
   delete[] m;

   return 0;
}
```



Exercise 3: Dynamic String Array Input and Output

- Write a C++ program that takes ‘n’ strings from the user and **dynamically allocates** an array to store those strings
- Show all the entered strings along with their corresponding indices

- Example run:

```
Enter the number of strings: 2
String 1: dynamic allocation
String 2: memory leaks

Output:
[0]: dynamic allocation
[1]: memory leaks
```